

Christopher Fernandez

Senior Data Engineer

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SUMMARY

I'm a Senior Data Engineer with 12+ years of experience building and optimizing ETL and ELT pipelines and data integration solutions across financial, healthcare, and telecommunications domains. Expert in cloud-based platforms including Azure Data Factory, Snowflake, and Azure Data Lake, as well as Teradata and Informatica PowerCenter, delivering scalable and high-performance data architectures. Skilled in data warehousing, batch processing, performance tuning, and automation, with a strong focus on data quality, security, and compliance. Proven ability to collaborate with cross-functional teams to translate complex business requirements into actionable insights and robust, end-to-end data solutions.

SKILLS

- **Cloud Platforms:**
Azure, AWS, Google Cloud Platform
 - **Azure Services:**
Data Factory, Data Lake Storage, Synapse Analytics, Analysis Services, Logic Apps, DevOps, Cosmos DB, SQL Database
 - **Big Data:**
Databricks, Apache Spark, PySpark, Delta Lake, Apache Kafka
 - **ETL Tools:**
Informatica PowerCenter, Informatica Cloud, SSIS, Azure Data Factory
 - **Databases:**
Snowflake, Teradata, SQL Server, Oracle, Azure SQL Database
 - **Programming:**
Python, T-SQL, SQL, PySpark
 - **Reporting:**
Power BI, SSRS
 - **DevOps:**
Git, GitHub, Azure DevOps, Terraform, CI/CD
 - **Other:**
Dynamics 365, Agile, BigQuery, Dataflow
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WORK HISTORY

Vise

New York, NY

Senior Data Engineer

Jan 2021 – Current

- Built and maintained data pipelines for an **investment management** platform, processing daily financial transactions and portfolio data to support reporting and analytics needs.
- Created data integration workflows in **Informatica Cloud** to move large financial datasets into **Azure Data Lake**, handling everything from initial setup to troubleshooting production issues.
- Developed **ETL pipelines** in **Databricks** using **PySpark** and **Apache Spark** to transform raw data from multiple sources into clean, queryable formats for business reporting.
- Wrote **Python** scripts with **Pandas** and **NumPy** to automate data extraction from APIs and flat files, handling data validation and error logging before loading into **Azure Data Lake**.
- Maintained legacy **SSIS** packages for core data warehouse loads while gradually migrating workloads to more modern cloud-based solutions.

- Pulled data from **Dynamics 365** using **Azure Data Factory** and **RESTful web services**, transforming it for use in downstream analytics and dashboards.
- Designed and built **ETL processes** to migrate on-premise databases to **Snowflake**, including data profiling, cleansing, and validation to maintain data quality.
- Implemented **Delta Lake** tables in **Databricks** to enable incremental data refreshes and maintain historical versions of key datasets in a **data lakehouse** architecture.
- Built transformation models using **dbt** to create reusable data models, implementing tests and documentation for better data quality and team collaboration.
- Set up data processing jobs in **BigQuery** and **Dataflow** to handle batch analytics workloads, optimizing queries for cost and performance.
- Built end-to-end data pipelines in **Microsoft Fabric**, orchestrating data movement from source systems into **Fabric Lakehouses** using Data Factory pipelines and Dataflow Gen2.
- Designed medallion architecture (**bronze/silver/gold layers**) in Fabric Lakehouses, implementing incremental loads and data quality checks at each layer using notebooks and pipelines.
- Created reusable data transformation logic in **Fabric notebooks** using **PySpark**, processing data stored in OneLake and writing results back to lakehouse tables for downstream consumption.
- Designed star schema **data warehouses** in **Azure Synapse Analytics**, creating dimension and fact tables to support **Power BI** dashboards and ad-hoc analysis.
- Built **Azure Functions** to handle event-driven data processing tasks and used **Azure Logic Apps** to automate notifications and workflow approvals.
- Set up **Kafka** streaming pipelines with **Event Hubs** to capture real-time transaction events, processing them through **Stream Analytics** for near real-time reporting.
- Developed **data lakehouse** solutions combining the flexibility of data lakes with the performance of data warehouses, enabling both analytics and **machine learning** workloads on the same platform.
- Built data pipelines to support **machine learning** models, preparing feature datasets and creating automated workflows to retrain models on fresh data.
- Implemented **MLOps** practices including model versioning, automated testing, and deployment pipelines to move ML models from development to production environments.
- Managed ETL deployments through **Azure DevOps** pipelines, writing YAML configurations for **CI/CD** and coordinating releases with the team.
- Configured **Snowflake** warehouse sizes, user roles, and access permissions while implementing monitoring and cost controls for the environment.
- Deployed infrastructure using **Terraform** to provision Azure resources like **SQL databases**, storage accounts, and **Databricks** workspaces as code.
- Containerized data processing applications with **Docker** and deployed them on **Kubernetes** clusters for better scalability and resource management.
- Used **Airflow** to orchestrate complex data workflows, managing dependencies between jobs and handling retries and alerts for failed tasks.
- Created **Power BI** reports and dashboards connected to **Azure SQL Database** and **Synapse**, working with business users to refine metrics and visualizations.
- Worked closely with analysts and data scientists to understand their data needs, translating requirements into pipeline specs and debugging data quality issues together.

Technologies: Azure (Data Factory, Databricks, SQL Database, Synapse Analytics, Functions, Logic Apps, Event Hubs, Stream Analytics, DevOps), AWS, Apache Spark, PySpark, Apache Kafka, Airflow, Informatica Cloud, SSIS, Snowflake, BigQuery, Dataflow, Python, Pandas, NumPy, T-SQL, Power BI, Dynamics 365, Docker, Kubernetes, Terraform, Delta Lake, dbt, MLOps, Machine Learning, Data Lakehouse, RESTful APIs

GeoWealth

Data Engineer

Chicago, IL

Sep 2016 – Dec 2020

- Worked on **financial services** projects, building data pipelines and analytics platforms to process transactional, portfolio, and banking data for reporting and analysis.
- Built **ETL workflows** using **Informatica PowerCenter**, creating mappings and transformations to move data between systems and apply business rules.

- Set up **data pipelines** connecting **Azure Synapse**, **Databricks**, and **Azure Data Lake Storage** to support our cloud analytics infrastructure.
- Created **Azure Data Factory** pipelines to pull data from various sources, using copy activities and data flows to move and transform data.
- Connected **Oracle databases** to our ETL processes using **SSIS**, **Informatica**, and **ADF**, making the data available for reporting.
- Wrote **PySpark** notebooks in **Azure Databricks** to transform and process data, storing results in **Delta Lake** tables for downstream use.
- Tuned **Spark jobs** in **Databricks** to run faster, cutting down processing times by about 30% through partitioning and caching strategies.
- Built pipelines in **BigQuery** and **Dataflow** to process large volumes of financial data for analytics and reporting needs.
- Worked with **Azure Synapse** and **Azure Analysis Services** to build data models and support business intelligence requirements.
- Set up **Azure Logic Apps** to automate recurring data tasks and trigger workflows based on specific events or schedules.
- Created **Python** scripts to check data quality, flag anomalies, and validate records before loading them into production tables.
- Handled day-to-day **Teradata** database tasks like creating tables, troubleshooting performance issues, and supporting user queries.
- Developed **SSRS** reports and **Power BI** dashboards that gave business users visibility into operations and key metrics.
- Optimized slow-running **SQL queries** and ETL jobs by adding indexes, rewriting logic, and adjusting data loading strategies.
- Used **Terraform** to script infrastructure setup and managed code changes through **Git** and **GitHub** for better team collaboration.
- Met regularly with business analysts and stakeholders to understand their data needs and figure out the best way to deliver solutions.

Technologies: Informatica PowerCenter, Azure (Data Factory, Synapse Analytics, Databricks, Data Lake Storage, Logic Apps, Analysis Services), Apache Spark, PySpark, Delta Lake, SSIS, SQL Server, SSRS, BigQuery, Dataflow, Teradata, Oracle, Python, Power BI, Terraform, Git, GitHub

Telnyx

Chicago, IL

ETL Engineer

May 2012 – Aug 2016

- Worked on **telecommunication** projects, building **ETL** pipelines and data solutions to process call records, network usage, and billing data.
- Designed and developed batch processing pipelines using **Azure Data Factory** for data ingestion from multiple sources.
- Automated ETL jobs in ADF using Event, Schedule, and Tumbling triggers to improve operational efficiency and reduce manual intervention.
- Built streaming data pipelines using Apache **Kafka** to process high-volume call records and network events in real-time.
- Managed **Teradata** environments using FastLoad, MultiLoad, and TPT for data integration across enterprise systems.
- Configured **TDWM/TASM** for workload management, performance tuning, and query optimization to ensure high system throughput.
- Designed **Azure SQL Database** and **SQL Server** schemas, stored procedures, and optimized queries to support operational analytics.
- Modeled **data warehouse** schemas using star and **snowflake** designs for optimized reporting and analytical workloads.
- Created **Power BI** dashboards to deliver network performance insights and operational reports to business stakeholders.

- Collaborated with cross-functional teams in **Agile** environment, creating technical documentation and translating requirements into ETL solutions.

Technologies: Azure, Apache Kafka, Teradata (FastLoad, MultiLoad, TPT, TDWM, TASM), Snowflake, SQL Server, Power BI, Python, Agile

EDUCATION

University of Illinois Chicago
Bachelor of Science

Chicago, IL
Sep 2008 – May 2012